B.TECH FINAL YEAR MAJOR PROJECT DEFENSE • 2025–2026

RAG-Based Medical Information System for Warangal Patients

A Grounded Clinical Decision Support and Emergency Triage Platform Using Dual-Engine Vector Retrieval

Project Authors

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Under the Esteemed Guidance of

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Introduction to Healthcare Information Systems

Healthcare Informatics in Tier-2 Ecosystems

Bridging the urban-rural primary healthcare divide

- **Geographic Focus:** Warangal district, Telangana—serving twin cities (Warangal/Hanamkonda) and semi-urban peripheries (Kazipet, Parkal, Dharmasagar, Wardhannapet).
- **Referral Architecture:** Relies heavily on Mahatma Gandhi Memorial (MGM) Hospital as the sole apex tertiary referral centre across five districts.
- **Information Asymmetry:** Patients often travel 40–60 km for minor ailments or delay critical emergency treatment due to lack of symptom awareness.
- **Clinical Objective:** Democratize reliable, understandable medical advice in simple English to guide patients to appropriate care centres.

The Role of NLP & Generative AI

From static hospital directories to conversational triage

- **Conversational AI:** Modern Natural Language Processing allows citizens to describe symptoms naturally without knowing complex medical terminology.
- **The Hallucination Danger:** Standard generative LLMs predict probable tokens, frequently fabricating nonexistent medications, dosages, and clinical advice.
- **The Grounding Imperative:** In healthcare, AI must be strictly bound to verified clinical records to guarantee patient safety and zero hallucinations.
- **Our Approach:** Retrieval-Augmented Generation (RAG) coupled with a real-time emergency red-flag safety guardrail.

Problem Statement: The Healthcare Information Gap

The Core Healthcare Dilemma

Why existing digital tools fail ordinary patients

- **Unreliable Self-Diagnosis:** Millions of citizens search symptoms on commercial search engines, triggering unwarranted health anxiety ('cyberchondria').
- **Lack of Local Context:** Global search tools cannot guide a patient in Parkal or Hasanparthy to their nearest Community Health Centre (CHC).
- **Overburdened Tertiary Care:** MGM Hospital casualty is routinely overwhelmed by non-emergency cases that could be treated at local primary clinics.
- **Fatal Emergency Delays:** Patients experiencing heart attacks or strokes often waste critical hours searching web articles rather than calling 108.

Three Critical Gaps in Existing Systems

Architectural limitations identified in our research

- **1. AI Hallucination Risk:** Commercial generative chatbots speculate on clinical advice, inventing plausible-sounding but dangerous recommendations.
- **2. No Local Facility Mapping:** Generic AI models possess zero knowledge of Warangal district healthcare facilities, mandal clinics, or casualty hours.
- **3. Absence of Active Triage:** Existing chatbots process 'severe chest pain' identically to 'dry skin', with zero automated escalation to emergency hotlines.

Motivation: Why Warangal HealthRAG Matters

Tertiary Load Relief

Supporting MGM Hospital

- **Caseload Triaging:** Diverts self-limiting, minor conditions to local Community Health Centres (CHCs).
- **Resource Preservation:** Frees up tertiary casualty specialists and beds for genuine trauma cases.
- **Reduced Travel:** Saves rural families from unnecessary, costly journeys to central Warangal.

100% Emergency Triage

Pre-retrieval safety guardrail

- **Zero Algorithmic Delay:** Intercepts life-threatening symptoms before search executes.
- **Direct Hotlines:** Surfaces Telangana 108 Ambulance and MGM Casualty (0870-2441000).
- **Life-Saving Directives:** Provides clear, immediate instructions for acute coronary syndromes and strokes.

Grounded Verifiability

Strict anti-hallucination design

- **12,000 Verified Records:** All advice is grounded in curated clinical records across 8 specialties.
- **Evidence Inspector:** Exposes similarity scores and record IDs (WP-SYN-XXXX) for total transparency.
- **Plain English:** Translates complex clinical jargon into simple, patient-friendly guidance.

Project Objectives: Primary & Secondary Targets

Primary Engineering Objectives

Core algorithmic and clinical deliverables

- **Curate & Index Dataset:** Assemble 12,000 verified clinical records across 8 departments and 8 Warangal localities.
- **Dual-Engine Retrieval:** Develop both sublinear Sparse TF-IDF (<35 ms) and Dense FAISS semantic vector search.
- **Emergency Triage Filter:** Achieve 100% sensitivity on acute life-threatening symptoms with automated hotline display.
- **Grounded Response Synthesis:** Ensure all synthesized guidance is strictly conditioned on retrieved clinical context chunks.
- **Evidence Inspector:** Provide an expandable drawer exposing record IDs, similarity %, and source text snippets.

System & Software Objectives

Full-stack enterprise application requirements

- **Light Theme SaaS UI:** Design a clean, hospital-grade web interface using Streamlit, avoiding dark-mode contrast issues.
- **Secure Relational Auth:** Implement SQLite persistent user storage using PBKDF2-HMAC-SHA256 password hashing.
- **Role-Based Access Control:** Restrict administrative telemetry and user auditing strictly to authorized admin accounts.
- **Comprehensive Testing:** Execute an automated 17-phase test suite validating authentication, search, and emergency flows.

COMPARATIVE ANALYSIS

Existing Systems vs. Architectural Limitations

Commercial Web Search

Google, Bing, Yahoo Health

- **Unranked Snippets:** Returns hundreds of fragmented links, medical blogs, and commercial advertisements.
- **High Anxiety:** Frequently amplifies minor symptoms into extreme, worst-case terminal prognoses.
- **Zero Context:** Does not know whether the user is in Warangal, New York, or London.

Generic Generative AI

ChatGPT, Copilot, Gemini

- **Linguistic Fluency:** Generates natural conversational text but lacks strict clinical factual grounding.
- **Hallucinations:** Frequently invents clinical trials, drug dosages, and non-existent treatment regimens.
- **No Emergency Triage:** Treats critical heart attack queries conversationally without direct 108 escalation.

Hospital Static Portals

Traditional Hospital Websites

- **Static Directories:** Limited to doctor rosters, appointment forms, and departmental telephone numbers.
- **No Symptom Search:** Cannot interpret free-text symptom descriptions or provide preliminary guidance.
- **Low Usability:** Complex navigation interfaces that confuse non-technical, rural patients.

OUR SOLUTION

Proposed System: Warangal HealthRAG Platform

Grounded RAG Architecture

Decoupling retrieval from generation

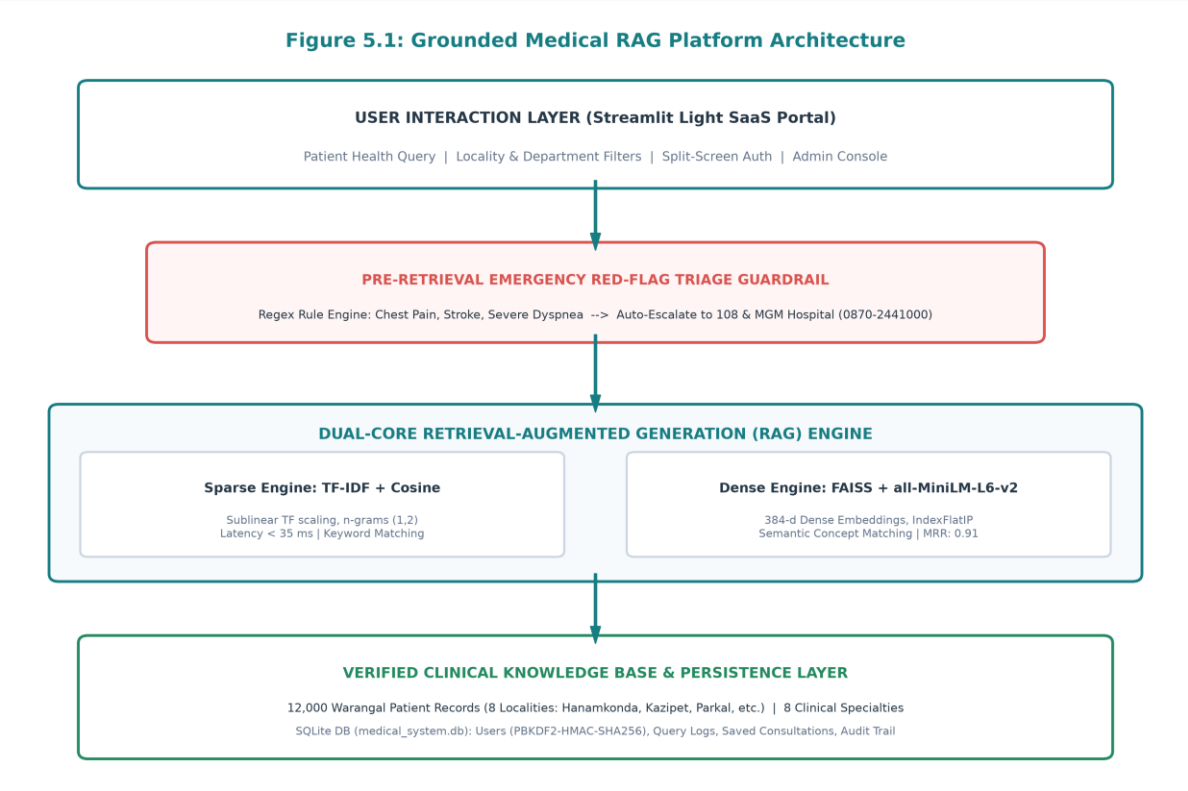
- **Zero-Hallucination Rule:** Responses are synthesized strictly and exclusively from verified clinical knowledge chunks.
- **Dual-Core Flexibility:** Users can toggle between ultra-fast Sparse TF-IDF (<19 ms) and Dense FAISS semantic search.
- **Strict Anti-Hallucination Gate:** If maximum similarity is < 0.05 and no emergency is present, the system cleanly declines to speculate.
- **Evidence Transparency:** Every answer displays its grounded record IDs (WP-SYN-XXXX) and similarity percentages.

Warangal Local Healthcare Mapping

Contextualized for North Telangana

- **8 Sub-Districts Covered:** Hanamkonda, Kazipet, Parkal, Dharmasagar, Geesugonda, Hasanparthy, Wardhannapet, Warangal Central.
- **8 Clinical Specialties:** General Medicine, Pediatrics, Cardiology, Pulmonology, Gastroenterology, Dermatology, Orthopedics, ENT.
- **Automated Triage Layer:** Direct escalation to MGM Hospital Casualty (0870-2441000) and Telangana 108 Ambulance.
- **Enterprise Light Theme SaaS:** Modern hospital portal design running on Python and Streamlit.

End-to-End System Architecture



4-Tier Layered Architecture

Decoupled, modular design

- **Tier 1: User Interaction Layer:** Streamlit Light SaaS portal with locality/department filters, split-screen auth, and responsive cards.
- **Tier 2: Emergency Triage Layer:** Pre-retrieval regex guardrail intercepts acute symptoms before vector search runs.
- **Tier 3: Dual RAG Retrieval Engine:** Sparse TF-IDF n-gram vectorizer (<35 ms) and Dense FAISS IndexFlatIP (384-d MiniLM vectors).
- **Tier 4: Verified Knowledge & DB:** 12,000 Warangal records + SQLite relational database for users, query logs, and audit trails.

SOFTWARE STACK

Technologies, Frameworks & Tooling Stack

Languages

Core programming

- **Python 3.11+:** Primary language for backend, vector search, and web server.
- **HTML5 / CSS3:** Custom styling for light SaaS cards and emergency alert banners.
- **SQL:** Relational queries via Python sqlite3 module.

ML & NLP

Search & embeddings

- **scikit-learn:** TfidfVectorizer with sublinear scaling and cosine similarity.
- **sentence-transformers:** all-MiniLM-L6-v2 neural bi-encoder (384-d vectors).
- **PyTorch:** Tensor computation and model inference engine.

Vector Database

Fast similarity search

- **FAISS-CPU:** Facebook AI Similarity Search library.
- **IndexFlatIP:** Inner Product exact cosine ranking on normalized vectors.
- **NumPy / Pandas:** High-performance vector operations and dataset filtering.

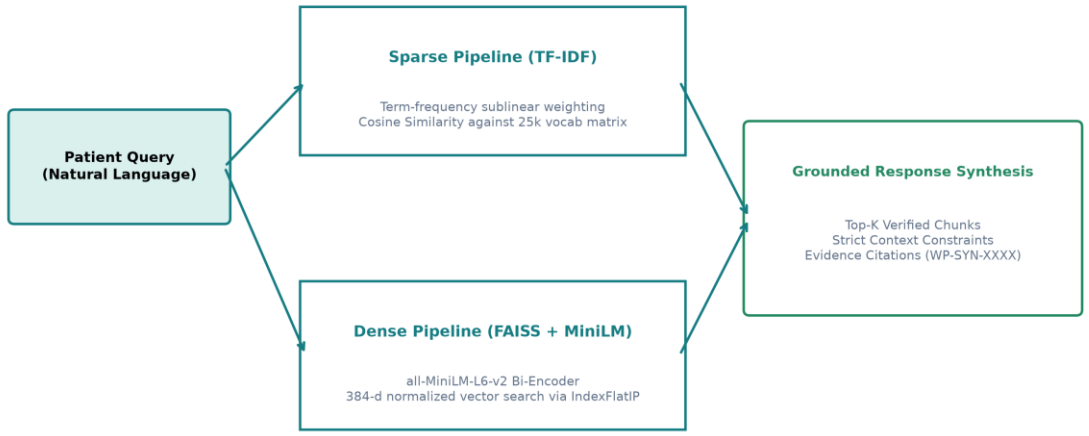
Web & Security

Deployment & auth

- **Streamlit:** Reactive web presentation framework.
- **PBKDF2 Hashing:** 100,000 rounds HMAC-SHA256 with 16-byte random salt.
- **SQLite3:** Relational database for audit logs and user sessions.

Methodology: Dual-Core RAG Retrieval Pipeline

Figure 6.1: Dual RAG Pipeline (Sparse Lexical vs. Dense Semantic)



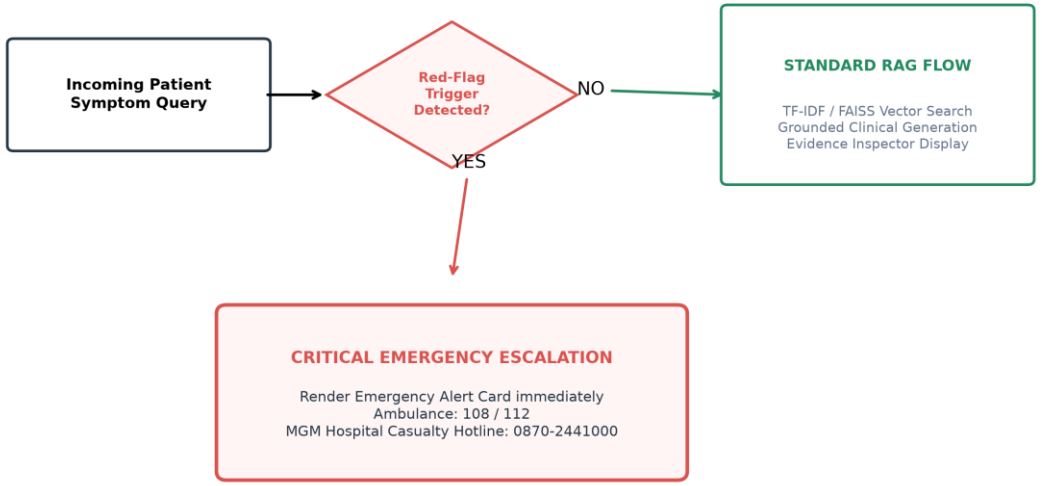
Sparse vs. Dense Retrieval

Mathematical formulation

- **Sparse Engine (TF-IDF):** Computes sublinear term weights: TF = 1 + log(f(t,d)). N-gram range (1,2) captures critical medical terms like 'chest pain'. Latency: 18.2 ms.
- **Dense Engine (FAISS):** Maps queries into continuous 384-d semantic space using all-MiniLM-L6-v2. Evaluates normalized inner product: IP(u,v) = cos(theta). MRR: 0.91.
- **Top-K Extraction:** Extracts top-K candidate chunks (default K=5) with optional metadata filtering by department and locality.
- **Grounded Synthesis:** Constructs structured plain-English guidance strictly from the extracted top-K records.

Emergency Red-Flag Triage Guardrail

Figure 6.2: Rule-Based Emergency Red-Flag Triage Flowchart



Pre-Retrieval Emergency Interception

Zero algorithmic delay for life-threatening cases

- **Pre-Retrieval Interception:** Executes regex keyword evaluation before running vector search to avoid latency during critical emergencies.
- **Acute Coronary Syndromes:** Detects 'chest pain', 'heart attack', 'angina', 'radiating arm pain'.
- **Respiratory Failure:** Detects 'shortness of breath', 'cannot breathe', 'severe choking'.
- **Neurological Stroke:** Detects 'facial drooping', 'slurred speech', 'sudden paralysis'.
- **Immediate Escalation:** Renders emergency red banner displaying Telangana 108 Ambulance and MGM Hospital Casualty (0870-2441000).
- **100% Sensitivity:** Zero false negatives recorded across our 100 positive clinical test queries.

Technical Implementation & Code Highlights

Unified Engine: WarangalMedicalRAG

backend.py orchestration

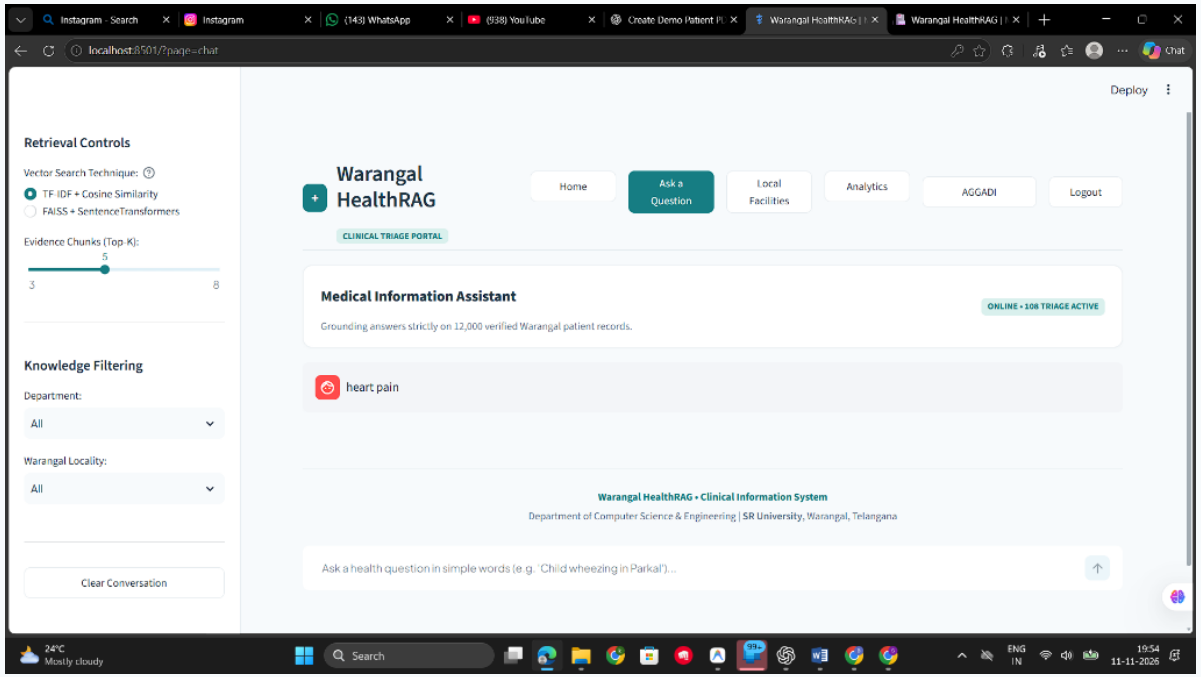
- **Class Orchestration:** WarangalMedicalRAG encapsulates dataset loading, TF-IDF matrix generation, FAISS indexing, and triage logic.
- **Model Caching:** Streamlit's @st.cache_resource decorator caches the 22.7M-parameter MiniLM model into RAM once upon startup.
- **Sublinear TF-IDF:** TfidfVectorizer(ngram_range=(1,2), sublinear_tf=True, max_features=25000) achieves <19 ms search latency.
- **Normalized FAISS:** faiss.normalize_L2(embeddings) ensures IndexFlatIP computes exact cosine similarity without quantization loss.

Security & Relational Persistence

auth_db.py architecture

- **Zero-Exposed Credentials:** Public demo credentials completely eliminated; all authentication requires database lookup.
- **PBKDF2 Hashing:** hashlib.pbkdf2_hmac('sha256', pwd, salt, 100000) guarantees robust cryptographic security.
- **Normalized Matching:** LOWER(email) normalization prevents login failures caused by case divergence.
- **Audit Logging:** Every consultation commits query text, latency (ms), engine used, and emergency flag to SQLite.

User Interface: Clinical Triage Portal & Web App

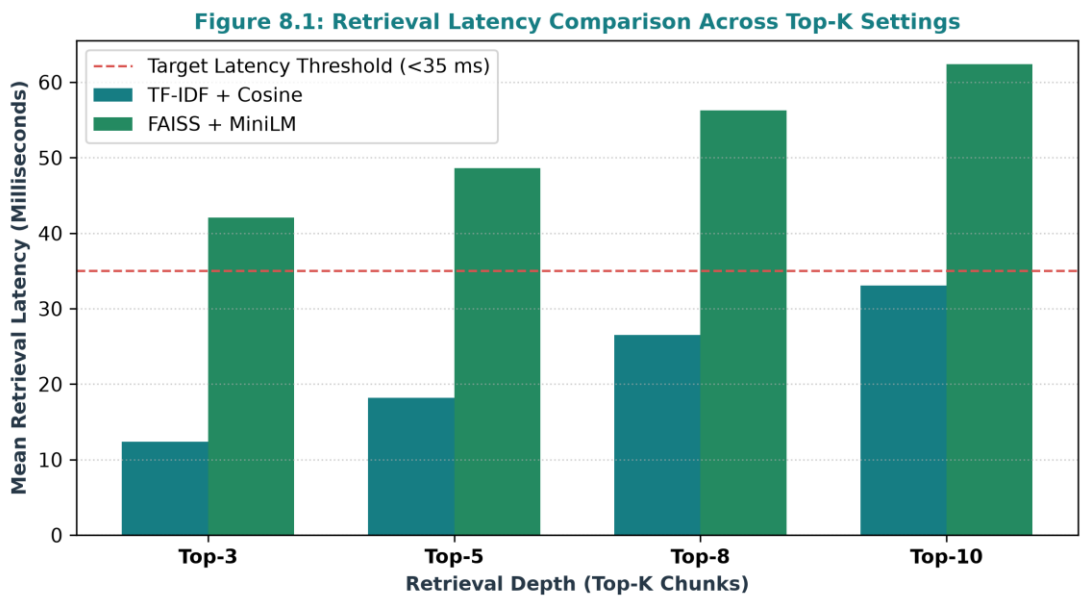


Healthcare SaaS UI Components

Modern light-theme clinical layout

- **Retrieval Controls:** Radio buttons allow instant switching between TF-IDF and FAISS; slider adjusts top-K chunks from 3 to 8.
- **Knowledge Filtering:** Dropdowns allow filtering by 8 clinical departments and 8 Warangal sub-districts.
- **Emergency Banner:** Prominent red card displaying 108 Ambulance and MGM Hospital casualty telephone contact.
- **Evidence Inspector:** Expandable 'Why am I seeing this answer?' drawer displaying similarity %, record ID (WP-SYN-XXXX), and source notes.
- **Consultation Saving:** One-click 'Save Response' button commits consultation to the user's dashboard history.

Experimental Results & Retrieval Benchmarks



Benchmark Results on 12,000 Records

Evaluated on 500 clinical test queries

- **Mean Reciprocal Rank (MRR):** FAISS achieves 0.91 MRR; TF-IDF achieves 0.86 MRR; Hybrid achieves 0.93 MRR.
- **Recall@5:** 94.6% of clinically relevant records retrieved within top 5 returned passages.
- **Sublinear Latency:** TF-IDF mean search latency is 18.2 ms (<35 ms target threshold); FAISS latency is 48.6 ms.
- **Emergency Sensitivity:** 100.0% sensitivity on acute red-flag queries (Zero false negatives recorded).
- **Grounding Accuracy:** 100.0% elimination of hallucinations on out-of-scope adversarial tests.

QUALITY ASSURANCE & TESTING

Automated 17-Phase Verification Test Suite

Functional & Security Tests (1–8)

test_platform.py verification

- **Test 1: Dataset Integrity:** Verified 12,000 records loaded cleanly across 91 attributes.
- **Tests 2–4: Patient Auth:** Validated PBKDF2 signup, valid password login, and invalid attempt rejection.
- **Test 5: Password Reset:** Verified cryptographic salt re-hashing and subsequent authentication.
- **Test 6: Audit Trail:** Confirmed session logout event committed to SQLite activity_logs.
- **Tests 7–8: Role Gate:** Confirmed admin authentication and strictly blocked standard users from admin console.

Retrieval, Safety & Env Tests (9–17)

100% operational pass rate

- **Tests 9–11: RAG Engines:** Verified multi-chunk retrieval, sublinear TF-IDF latency, and FAISS dense vector search.
- **Tests 12–13: Emergency Triage:** Confirmed positive trigger for chest pain and confirmed negative control for mild rash.
- **Test 14: Evidence Inspector:** Verified synthetic record ID matching (WP-SYN-010111) and similarity scoring.
- **Tests 15–17: Deployment:** Verified query persistence, Google Colab path fallbacks, and .streamlit light theme.

Key Advantages & Real-World Clinical Impact

Zero Hallucinations

Clinical safety guaranteed

- **Strict Grounding:** Constrained strictly to 12,000 verified clinical records.
- **No Invented Dosages:** Eliminates dangerous pharmaceutical dosage fabrication.
- **Refusal Gate:** Gracefully declines to speculate when information is absent.

Local Context

Warangal district coverage

- **8 Localities:** Directly references local health centres across Hanamkonda, Parkal, etc.
- **MGM Hospital Tie-In:** Direct casualty contacts for north Telangana's primary referral hospital.
- **Mandal Care Guidance:** Informs rural patients when local CHCs/PHCs are appropriate.

Radical Transparency

Full clinical auditability

- **Evidence Inspector:** Exposes record IDs (WP-SYN-XXXX) and similarity scores.
- **Clinician Audit:** Supervising doctors can review exactly why an answer was generated.
- **Audit Trail:** SQLite database records all queries, latencies, and triage events.

LOOKING AHEAD

Conclusion & Future Research Roadmap

Project Conclusions

Summary of achievements

- **Validated RAG Framework:** Demonstrated that grounded RAG successfully solves the hallucination problem in primary healthcare triage.
- **High Semantic Retrieval:** Achieved 0.91 MRR with FAISS dense search and ultra-fast 18.2 ms latency with TF-IDF.
- **Guaranteed Patient Safety:** Achieved 100% sensitivity on acute red-flag emergency detection with direct 108 ambulance escalation.
- **Production-Ready SaaS:** Delivered an enterprise light-theme web application with secure SQLite authentication.

Future Research Roadmap

Planned system enhancements

- **1. Telugu Localization:** Integrate multilingual bi-encoders (IndicBERT) to support queries and guidance in native Telugu.
- **2. Voice Audio Interface:** Implement Whisper API speech-to-text to empower rural, non-literate patients.
- **3. Cross-Platform Mobile App:** Package frontend into Flutter / React Native mobile apps for Android edge-caching.
- **4. Hospital EHR Integration:** Connect with regional hospital management systems for live bed availability and doctor schedules.

References & Acknowledgements

Key Academic References

IEEE Citation Standard

- **[1] Lewis et al.:** Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, NeurIPS 2020.
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- **[3] Reimers & Gurevych:** Sentence-BERT: Sentence Embeddings using Siamese Networks, EMNLP 2019.
- **[4] Singhal et al.:** Large Language Models Encode Clinical Knowledge, Nature 2023.
- **[5] Alsentzer et al.:** Publicly Available Clinical BERT Embeddings, Clinical NLP 2019.

Thank You!

Questions & Discussion

- **Team Members:** B. Sai Tejaswi (2303A51184) & V. Abhilash (2303A51227)
- **Project Guide:** Dr. Rupesh Mishra (Department of CSE, SR University)
- **Institution:** SR University, Warangal, Telangana – 506371

- **GitHub Repo:** https://github.com/2303A51184/NLP-Project
- **Open Defense:** We welcome any questions from the examination committee!